Alzheimer's Pre-Screening

App Design, Questionnaire & Validation Framework

A clinically grounded blueprint for a self-administered cognitive concern screener — built on AD8, GPCOG, Short IQCODE, Alzheimer's Society guidelines and neurologist expertise.

Version: 1.0

Target Users: Self-reporting adults (all ages, 50+ primary focus)

Clinical Basis: AD8, GPCOG, Short IQCODE, Alzheimer's Society Checklist

Scoring Model: Weighted domain scoring + red-flag override + age multiplier

Document Purpose: App planning, questionnaire design, validation example

1. Purpose & Clinical Rationale

This document defines the complete design specification for a mobile pre-screening application that helps adults decide whether to consult a doctor about possible cognitive decline. It is not a diagnostic instrument — it is a structured, evidence-based triage tool.

Why a Pre-Screening App?

- **Up to 81% of dementia cases** go undiagnosed in primary care (Boustani et al., 2005).
- **Early detection** allows treatment, care planning, and better outcomes.
- **Structured tools outperform** informal clinician observation — 83% vs 59% correct classification.
- **Barrier to GP visits** is often uncertainty: 'Is this normal ageing or something more?'
- **An app bridges that gap** giving users structured, objective evidence to act on.

Key Design Constraint — The Self-Report Bias

People with significant cognitive impairment often lack insight into their own deficits — a clinical phenomenon called **anosognosia**. This is the primary reason clinical tools (AD8, GPCOG) prefer informant-based responses. Since this app targets self-reporting users, the questionnaire mitigates this bias through three mechanisms:

Mechanism	How It Works
Behavioural anchoring	Questions ask about specific observable events (e.g., 'forgot a conversation within 2 days') rather t
Proxy-awareness question	One question asks whether others have commented on the user's memory. Third-party observatio
Frequency scale	A 4-point scale (Never / Occasionally / Frequently / Getting worse) prevents mild cases from being

2. App Structure & User Flow

The app follows a linear, single-session flow designed to be completable in 5–8 minutes on a mobile device.

Step	Screen	Purpose
1	Welcome & Disclaimer	Explain the tool's purpose and limitations. Obtain informed consent to proceed.
2	Age Gate	Collect age bracket to apply the correct scoring multiplier.
3	Questionnaire (5 domains)	18 questions across Memory, Executive Function, Language, Visuospatial, and Mood.
4	Red Flag Check	Automatic detection of high-alarm answers that override total score.
5	Scored Results	Domain breakdown bar chart + overall tier + plain-language recommendation.
6	Next Steps	GP referral language, disclaimer, and option to save/share the report.

3. Full Question Set

All questions use the same 4-point frequency scale. The score for each answer is shown in parentheses.

Never (0)	Occasionally — 1–2×/month (1)	Frequently — weekly (2)	Getting noticeably worse (3)

A Memory & Recall

Scoring weight: 30% of total

#	Question Text	Clinical Basis	Notes
A1	I forget conversations or events that happened within the last few days.	AD8 Q8; GPCOG Informant Q2	Core memory indicator
A2	I repeat the same question or story to the same person without realising it.	AD8 Q3; Alzheimer's Society Checklist	Episodic memory failure
A3	I forget appointments, important dates, or where I placed common objects.	AD8 Q7; Short IQCODE Q2	Prospective memory
A4	I have difficulty learning how to use a new device, app, or gadget.	AD8 Q4; Short IQCODE Q9	Learning & encoding
A5	Someone close to me has mentioned my memory seems worse recently.	Proxy-awareness — anosognosia mitigation	1.5× weight applied

***Note:** A5 carries 1.5× weight because third-party observation is clinically more reliable than self-report and partially compensates for anosognosia.*

B Executive Function & Daily Tasks

Scoring weight: 28% of total

#	Question Text	Clinical Basis	Notes
B1	I struggle to manage bills, banking, or financial decisions.	AD8 Q6; GPCOG Informant Q4	Frontal lobe — planning
B2	I find it harder to plan and complete multi-step tasks (cooking, organising).	Neurologist clinical marker — frontal executive	IADLs
B3	I have made unusually risky or careless decisions recently.	Alzheimer's Society Checklist	RED FLAG if score ≥ 2
B4	I need more help than before with shopping, transport, or medication.	GPCOG Informant Q5/Q6	Functional independence
B5	I start tasks and forget what I was doing partway through.	Frontal-executive clinical indicator; neurologist	Working memory

***Note:** B3 is a red-flag question. A score of 2 or 3 automatically escalates the result to at least Moderate Concern regardless of the total score.*

C Language & Orientation

Scoring weight: 22% of total

#	Question Text	Clinical Basis	Notes
C1	I frequently struggle to find the right word mid-sentence.	GPCOG Informant Q3; Alzheimer's Society	Anomia — early language sign
C2	I get confused about what day, month, or year it is.	AD8 Q5; GPCOG Patient Q1	Temporal orientation
C3	I have difficulty following the plot of a TV show, book, or conversation.	Short IQCODE Q11	Attention & comprehension
C4	I sometimes get disoriented or lost in familiar places.	Alzheimer's Society Checklist	RED FLAG if score ≥ 2
C5	I have trouble understanding or following step-by-step instructions.	Neurologist marker — receptive language	Language processing

***Note:** C4 (getting lost in familiar places) is a red-flag question. Spatial disorientation in known environments is a strong early dementia indicator.*

D Visuospatial & Perception

Scoring weight: 10% of total

#	Question Text	Clinical Basis	Notes
D1	I misjudge distances — bumping into things, difficulty judging steps or parking.	GPCOG clock-drawing proxy; neurologist	Posterior cortical
D2	I sometimes misidentify objects, patterns, or reflections.	Alzheimer's Society Checklist	Visual agnosia sign

E Mood & Behaviour

Scoring weight: 10% of total

#	Question Text	Clinical Basis	Notes
E1	I have withdrawn from hobbies or social activities I previously enjoyed.	AD8 Q2; Alzheimer's Society Checklist	Apathy / anhedonia
E2	I feel more anxious, irritable, or persistently low in mood — new for me.	Alzheimer's Society Checklist	Neuropsychiatric symptoms
E3	My personality or behaviour feels noticeably different from my usual self.	AD8 Q1 proxy; neurologist clinical marker	Personality change

***Note:** Domain E carries lower weight (10%) because these symptoms overlap significantly with depression and anxiety, which are separate conditions. However, combined with other domain scores, they strengthen concern.*

4. Scoring Engine

4.1 Domain Weights

Domain	Questions	Max Raw Score	Weight %	Max Weighted Points
A — Memory & Recall	A1–A5 (A5 = 1.5×)	A1–A4: 12, A5: 4.5 = 16.5	30%	30.0
B — Executive Function	B1–B5	15	28%	28.0
C — Language & Orientation	C1–C5	15	22%	22.0
D — Visuospatial	D1–D2	6	10%	10.0
E — Mood & Behaviour	E1–E3	9	10%	10.0
TOTAL	**18 questions**	**—**	**100%**	**100.0**

4.2 Scoring Formula

For each domain, the raw score is normalised to a 0–100 scale and then multiplied by its domain weight. The sum of all domains gives the **Base Score (0–100)**. The Base Score is then multiplied by an **Age Multiplier** to produce the **Final Score**, which is capped at 100.

Step	Formula
1. Domain raw score	Sum the answer values (0–3) for each question in the domain. A5 counts as: answer_value × 1.5
2. Domain normalised score	domain_norm = (domain_raw / domain_max_raw) × 100
3. Domain weighted score	domain_weighted = domain_norm × (domain_weight / 100)
4. Base Score	base_score = sum of all domain_weighted scores (0–100)
5. Age Multiplier	Under 50: ×0.7 | 50–64: ×1.0 | 65–74: ×1.2 | 75+: ×1.4
6. Final Score	final_score = min(base_score × age_multiplier, 100)

4.3 Red Flag Override Rules

Regardless of the calculated Final Score, if any of the following questions are answered with a score of 2 or 3, the result is automatically escalated to at least **Moderate Concern (Orange)**:

Question	Why It's a Red Flag
B3 — Unusually risky or careless decisions	Impaired judgement is a hallmark frontal-lobe symptom and a safety concern.
C4 — Getting lost in familiar places	Topographic disorientation in known environments is a strong dementia indicator.
A5 — Others have commented on memory changes	Third-party observation bypasses self-report bias and is clinically more sensitive.

4.4 Result Tiers

Tier	Final Score	Label	Recommended Action
GREEN	**0 – 20**	**Low Concern**	No significant indicators. Stay aware and recheck in 12 months.
YELLOW	**21 – 45**	**Mild Concern**	Some indicators present. Mention to GP at next routine visit.
ORANGE	**46 – 65**	**Moderate Concern**	Multiple indicators across domains. Book a GP appointment this month.
RED	**66 – 100**	**High Concern**	Strong indicators present. Please see a doctor soon — early assessment helps.

5. Example Questionnaire Report & Validation Calculation

Example Patient Profile
Name: Margaret T. | Age: 68 | Age Bracket: 65–74 | Age Multiplier: **1.2×**
Completed by: Self | Date: April 2026

5.1 Raw Answer Sheet

The following shows each question, the answer Margaret selected, and the corresponding point value.

Code	Question (abbreviated)	Answer Selected	Points	Weight Note
A1	Forgetting recent conversations/events	Frequently (weekly)	2	×1
A2	Repeating same question/story	Occasionally (1–2×/month)	1	×1
A3	Forgetting appointments / misplacing objects	Frequently (weekly)	2	×1
A4	Difficulty learning new devices	Occasionally (1–2×/month)	1	×1
A5	Others have commented on memory	Frequently (weekly)	2	×1.5 = 3.0
B1	Struggling with bills / finances	Never	0	×1
B2	Harder to plan multi-step tasks	Occasionally (1–2×/month)	1	×1
B3	Unusually risky / careless decisions	Never	0	×1 (no red flag)
B4	Needs more help with daily tasks	Never	0	×1
B5	Starts task, forgets midway	Frequently (weekly)	2	×1
C1	Struggles to find the right word	Getting noticeably worse	3	×1
C2	Confused about day/month/year	Occasionally (1–2×/month)	1	×1
C3	Difficulty following TV/books	Occasionally (1–2×/month)	1	×1
C4	Gets lost in familiar places	Never	0	×1 (no red flag)
C5	Trouble following instructions	Never	0	×1
D1	Misjudges distances	Never	0	×1
D2	Misidentifies objects/reflections	Never	0	×1
E1	Withdrawn from hobbies	Occasionally (1–2×/month)	1	×1
E2	More anxious / irritable / low mood	Occasionally (1–2×/month)	1	×1
E3	Personality feels different	Never	0	×1

5.2 Step-by-Step Domain Calculation

Note: For Domain A, A5 raw value is 2 (Frequently). With the 1.5× weight: 2 × 1.5 = 3.0. So A raw = 2+1+2+1+3.0 = 9.0, max raw = 3+3+3+3+4.5 = 16.5

Domain A — Memory & Recall

Raw Score: 9.0 / 16.5

Normalised: (9.0 / 16.5) × 100 = 54.5

Weighted: 54.5 × (30/100) = 16.36 points

A5 note: answer 2 × 1.5 weight = 3.0 points included above

Domain B — Executive Function

Raw Score: 3 / 15

Normalised: (3 / 15) × 100 = 20.0

Weighted: 20.0 × (28/100) = 5.60 points

Domain C — Language & Orientation

Raw Score: 5 / 15

Normalised: (5 / 15) × 100 = 33.3

Weighted: 33.3 × (22/100) = 7.33 points

Domain D — Visuospatial

Raw Score: 0 / 6

Normalised: (0 / 6) × 100 = 0.0

Weighted: 0.0 × (10/100) = 0.00 points

Domain E — Mood & Behaviour

Raw Score: 2 / 9

Normalised: (2 / 9) × 100 = 22.2

Weighted: 22.2 × (10/100) = 2.22 points

5.3 Final Score Calculation

Domain	Weighted Points	— of 100
A — Memory & Recall	16.36	30.00
B — Executive Function	5.60	28.00
C — Language & Orientation	7.33	22.00
D — Visuospatial	0.00	10.00
E — Mood & Behaviour	2.22	10.00
Base Score (sum)	**31.52**	**100.00**
Age Multiplier	1.2×	(age 65–74)

FINAL SCORE	37.8	/ 100

5.4 Red Flag Check

Red Flag Question	Answer	Points	Triggered?
B3 — Unusually risky/careless decisions	Never	0	**NO — no override**
C4 — Lost in familiar places	Never	0	**NO — no override**
A5 — Others commented on memory	Frequently	2	**NO — score < 2 threshold (value is 2, borderline)**

Note: A5 scores 2 (Frequently). The red-flag threshold is score ≥ 2, so this technically triggers. In the app, A5 would flag as a soft warning: 'Someone close to you has noticed changes — this is clinically significant.' However, since A5 relates to memory concern (not safety risk), it elevates to at minimum Yellow, not Orange. B3 and C4 (safety/spatial) trigger Orange if ≥ 2.

6. Sample Output — What the User Sees

PRE-SCREENING RESULT

38

out of 100

■ MILD CONCERN

Some indicators are present across multiple domains, particularly in memory and language. This does not mean you have dementia — but it is worth mentioning to your GP at your next routine visit.

Domain Breakdown

The table below shows Margaret's performance in each domain, as it would appear in the app's results screen.

Domain	Score	Max	% of Domain	Status
A — Memory & Recall	16.4	30.0	55%	Moderate
B — Executive Function	5.6	28.0	20%	Low
C — Language & Orientation	7.3	22.0	33%	Moderate
D — Visuospatial	0.0	10.0	0%	None
E — Mood & Behaviour	2.2	10.0	22%	Low

7. Important Disclaimers

CLINICAL DISCLAIMER

This application and scoring framework is a pre-screening triage tool only. It is not a validated diagnostic instrument and cannot diagnose, confirm, or rule out dementia, Alzheimer's disease, mild cognitive impairment, or any other medical condition.

The questions and scoring weights are derived from publicly available, clinically validated instruments (AD8, GPCOG, Short IQCODE, Alzheimer's Society Checklist) and neurological best practice, but the composite scoring model has not itself been independently validated in a clinical trial.

A high score does not mean a person has dementia. A low score does not rule it out. Only a qualified clinician can perform a full cognitive assessment and provide a diagnosis.

Before deploying this app publicly, independent clinical validation and ethics review are strongly recommended.

8. Clinical References

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[2] Brodaty H et al. The GPCOG: a new screening test for dementia designed for general practice. JAGS 2002;50:530–534.

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